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1.

$$\begin{cases} 10x - y + 2z = 6 \\ -x + 11y - z = 22 \\ 2x - y + 10z = -10 \end{cases}$$

forma matricial:

$$Ax = b$$

$$A = \begin{bmatrix} 10 & -1 & 2 \\ -1 & 11 & -1 \\ 2 & -1 & 10 \end{bmatrix} \qquad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \qquad b = \begin{bmatrix} 6 \\ 22 \\ -10 \end{bmatrix}$$

sistema matricial completo:

$$\begin{bmatrix} 10 & -1 & 2 \\ -1 & 11 & -1 \\ 2 & -1 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 22 \\ -10 \end{bmatrix}$$

2.

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

1° linha:

$|10| > |-1| + |2|$

$10 > 1 + 2$

$10 > 3$ ✓

2° linha:

$|11| > |-1| + |-1|$

$11 > 1 + 1$

$11 > 2$ ✓

3° linha:

$|10| > |2| + |-1|$

$10 > 2 + 1$

$10 > 3$ ✓

A matriz A é estritamente diagonalmente por linhas, pois as três condições são satisfeitas

3.

$10x - y + 2z = 6$ $\qquad -x + 11y - z = 22$ $\qquad 2x - y + 10z = 10$

$10x = 6 + y - 2z$ $\qquad 11y = 22 + x + z$ $\qquad 10z = -10 - 2x + y$

$$\boxed{x = \frac{6 + y - 2z}{10}} \qquad \boxed{y = \frac{22 + x + z}{11}} \qquad \boxed{z = \frac{-10 - 2x + y}{10}}$$

$$x^{(k)} = \frac{6 + y^{(k-1)} - 2z^{(k-1)}}{10} \qquad y^{(k)} = \frac{22 + x^{(k)} + z^{(k-1)}}{11} \qquad z^{(k)} = \frac{-10 - 2x^{(k)} + y^{(k)}}{10}$$

4.

$$x^{(0)} = (0,0,0)$$

Logo:

$$x^{(0)} = 0 \;,\; y^{(0)} = 0 \;,\; z^{(0)} = 0$$

1° interação:

$$x^{(1)} = \frac{6 + y^{(0)} - 2z^{(0)}}{10} \Rightarrow x^{(1)} = \frac{6 + 0 - 2(0)}{10} = \frac{6}{10} \Rightarrow \boxed{x^{(1)} = 0{,}6}$$

$$y^{(1)} = \frac{22 + x^{(1)} + z^{(0)}}{11} \Rightarrow y^{(1)} = \frac{22 + 0{,}6 + 0}{11} = \frac{22{,}6}{11} \Rightarrow \boxed{y^{(1)} \approx 2{,}054545454}$$

$$z^{(1)} = \frac{-10 - 2x^{(1)} + y^{(1)}}{10} \Rightarrow z^{(1)} = \frac{-10 - 2(0{,}6) + 2{,}054545454}{10} = \frac{-9{,}145454545}{10} \Rightarrow \boxed{z^{(1)} \approx -0{,}914545454}$$

2° iteração :

$$x^{(2)} = \frac{6 + y^{(1)} - 2z^{(1)}}{10} \Rightarrow x^{(2)} = \frac{6 + 2,05454545 - 2(-0,91454545)}{10} \Rightarrow \boxed{x^{(2)} \approx 0,98836364}$$

$$y^{(2)} = \frac{22 + x^{(2)} + z^{(1)}}{11} \Rightarrow \frac{22 + 0,98836364 - 0,91454545}{11} \Rightarrow \boxed{y^{(2)} \approx 2,00671074}$$

$$z^{(2)} = \frac{-10 - 2x^{(2)} + y^{(2)}}{10} \Rightarrow \frac{-10 - 2(0,98836364) + 2,00671074}{10} \Rightarrow \boxed{z^{(2)} \approx -0,99700165}$$

3° iteração :

$$x^{(3)} = \frac{6 + y^{(2)} - 2z^{(2)}}{10} \Rightarrow \frac{6 + 2,00671074 - 2(0,99700165)}{10} \Rightarrow \boxed{x^{(3)} \approx 1,00007140}$$

$$y^{(3)} = \frac{22 + x^{(3)} + z^{(2)}}{11} \Rightarrow \frac{22 + 1,00007140 - 0,99700165}{11} \Rightarrow \boxed{y^{(3)} \approx 2,00027907}$$

$$z^{(3)} = \frac{-10 - 2x^{(3)} + y^{(3)}}{10} \Rightarrow \frac{-10 - 2(1,00007140) + 2,00027907}{10} \Rightarrow \boxed{z^{(3)} \approx -0,99998637}$$

5.

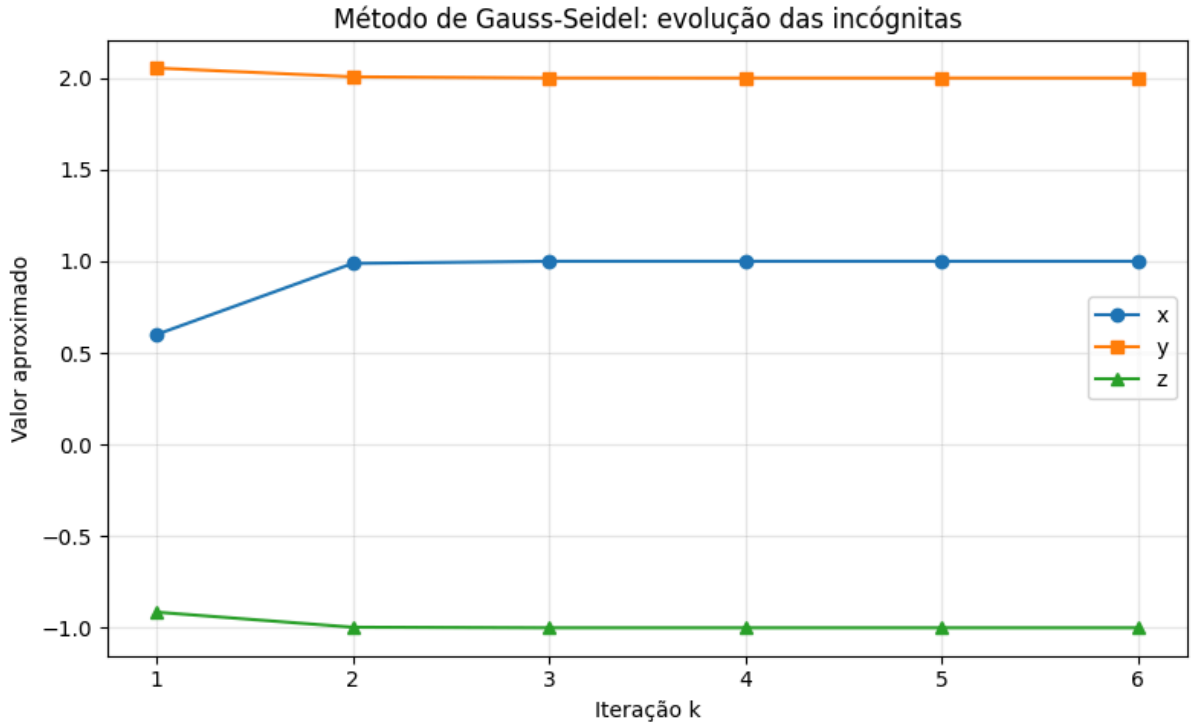
[tarefaGauss_celine.ipynb - Colab]

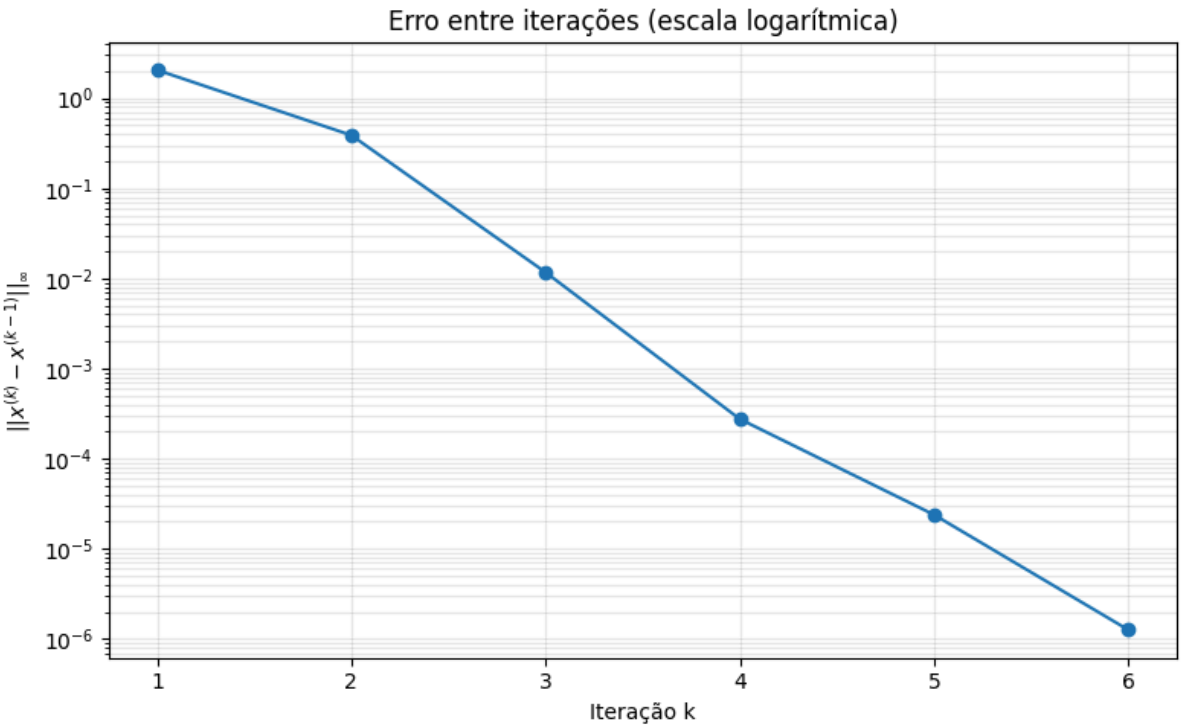
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Número de iterações: 6
Solução aproximada (x, y, z): [ 1.00000003  1.99999998 -1.00000001]
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6.

k	$x^{(k)}$	$y^{(k)}$	$z^{(k)}$	E_k
1	0,60000000	2,05454545	−0,91454545	2,05454545
2	0,98836364	2,00671074	−0,99700165	0,38836364
3	1,00007140	2,00027907	−0,99998637	0,01170777
4	1,00002518	2,00000353	−1,00000468	0,00027554
5	1,00000129	1,99999969	−1,00000029	0,00002389
6	1,00000003	1,99999998	−1,00000001	0,00000126

7.




Erro entre iterações (escala logarítmica)
$||x^{(k)} - x^{(k-1)}||_\infty$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
1
2
3
4
5
6
Iteração k